

## Class 15: Symmetric Groups

### Goals

1. Understand permutations as functions.
2. Understand array notation and cycle notation.
3. Calculate the number of permutations in  $S_n$ .

Let  $S_3$  be the set of all functions with domain  $\{1,2,3\}$  and range  $\{1,2,3\}$ . List all of the distinct functions in  $S_3$ .

| $\chi$ | $\xi(x)$ | $\alpha(x)$ |  |  |  |
|--------|----------|-------------|--|--|--|
| 1      | 1        | 1           |  |  |  |
| 2      | 2        | 3           |  |  |  |
| 3      | 3        | 2           |  |  |  |

Each function can be thought of as an arrangement of the numbers 1, 2, 3. These functions are one-to-one and onto from the set  $\{1,2,3\}$  to itself. Such functions are called \_\_\_\_\_ on the set  $\{1,2,3\}$ .

Big idea: The set of all permutations on the set  $\{1,2,\dots, n\}$  forms a group.

Array notation: Instead of writing a permutation in  $S_n$  using a vertical table of x-values and function values, we can use a horizontal two-dimensional array.

Examples in  $S_3$

$$\varepsilon = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} \quad \alpha = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix} \quad \beta = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\gamma =$$

$$\delta =$$

$$\varphi =$$

Explore the group operation

$$\varepsilon \circ \alpha = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} =$$

$$\varepsilon \circ \beta = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} =$$

$$\beta \circ \varepsilon = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} =$$

$$\alpha \circ \alpha = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} =$$

$$\alpha \circ \beta = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} =$$

$$\beta \circ \alpha = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} =$$

Make a complete Cayley table for  $S_3$ .

| $\circ$       | $\varepsilon$ | $\alpha$             | $\beta$ | $\gamma$ | $\delta$ | $\varphi$ |
|---------------|---------------|----------------------|---------|----------|----------|-----------|
| $\varepsilon$ |               |                      |         |          |          |           |
| $\alpha$      |               | $\alpha \circ \beta$ |         |          |          |           |
| $\beta$       |               | $\beta \circ \alpha$ |         |          |          |           |
| $\gamma$      |               |                      |         |          |          |           |
| $\delta$      |               |                      |         |          |          |           |
| $\varphi$     |               |                      |         |          |          |           |

The number of elements in a group is called \_\_\_\_\_

How many permutations are there in  $S_3$ ,  $S_4$ , or  $S_n$ ?

\_\_\_\_\_  $\circ$  \_\_\_\_\_

Cycle notation is a more efficient way to represent permutations for larger values of  $n$ .

Example:  $\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 5 & 6 & 3 & 1 \end{bmatrix}$

Let  $\alpha = (1246)(35)$  and  $\beta = (135)(246)$ .

$$\alpha \circ \beta =$$

Standard cycle form

Cycles are disjoint

Smallest numbers first.

Leading numbers in increasing order

Don't list fixed numbers

Exception for the identity permutation  $\xi = (1)$

Practice: Write each permutation in standard cycle notation form.

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 3 & 5 & 7 & 2 & 4 & 6 \end{bmatrix}$$

$$\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 6 & 5 & 4 & 3 & 2 & 1 \end{bmatrix}$$

$\alpha =$

$\beta =$

$$\gamma = \alpha \circ \beta =$$

$$\gamma^{-1} =$$

$$\gamma^2 =$$

List all distinct powers of  $\gamma$